

PnL Attribution & Desk Diagnostics

Reading PnL Like a Model Engineer

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Key Observations

- **PnL is a diagnostic signal, not a number:** It tells you which model assumption broke first
- **Attribution is hypothesis testing, not bookkeeping:** You're testing whether your model still describes reality
- **Residual PnL is the only honest feedback loop:** Everything else is model self-consistency
- **Clean vs Dirty PnL split is the first sanity check:** If you don't separate them, you'll misattribute model failure to market moves
- **Correlation is conditional, not structural:** Calm-market correlations are fiction; stress correlations are truth
- **Greeks are local derivatives of a wrong model:** Treat them as approximations, not facts
- **Reserves are for uncertainty, not losses:** If reserves are static, your understanding is outdated

Module 0: How to Use This Guide (Critical Framing)

Module Overview

What this module is about: This guide teaches you to think like a Model Engineer of PnL, not a bookkeeper. Every unexplained dollar is a clue to a broken assumption.

Why this matters: Most quants can price. Few can debug. On a trading desk, debugging pays your salary. Pricing is just the entry fee.

What confusion this clears:

- Why PnL attribution is model validation, not arithmetic
- Why residual is sacred and Greeks are suspect
- Why "explain 95% of PnL" is failure, not success

After this module, you should be able to answer: "What is the difference between a PnL report and a PnL diagnosis?"

0.1 Rules of Engagement

Warning Sign

Rule 1: Never trust first-pass Greeks

Greeks are computed from a model that is already wrong. They are the model's self-report, not independent verification.

Warning Sign

Rule 2: Always assume one assumption is broken

Your job is not to prove the model right. It's to find which assumption failed and how much it cost you.

Warning Sign

Rule 3: Market moves don't explain PnL models do

A 2% spot move explains nothing. The question is: did your delta predict the PnL from that 2% move? If not, your delta is wrong.

- Why calm-market correlation calibration is useless
- Why cross-gamma PnL surprises everyone
- How to detect correlation concentration before it explodes

After this module, you should be able to answer:

“Your 10-asset book is diversified at $\rho = 0.3$. Why did it lose \$5M in one day?”

6.1 Correlation Is Conditional, Not Structural

Core Formula

$$\rho_{\text{calm}} = 0.3 \xrightarrow{\text{Stress}} \rho_{\text{crisis}} = 0.95$$

The math: Correlation is a conditional expectation:

$$\rho_{XY} = \frac{\mathbb{E}[(X - \mu_X)(Y - \mu_Y) | \text{calm}]}{\sigma_X \sigma_Y}$$

The problem: Stress is a different conditioning set.

Practitioner Insight

The Desk Wisdom:

Never hedge correlation risk using ρ from the last 3 years of data. That ρ assumes the future looks like the past. In crisis, the past is irrelevant. Use $\rho = 0.8$ as a stress floor.

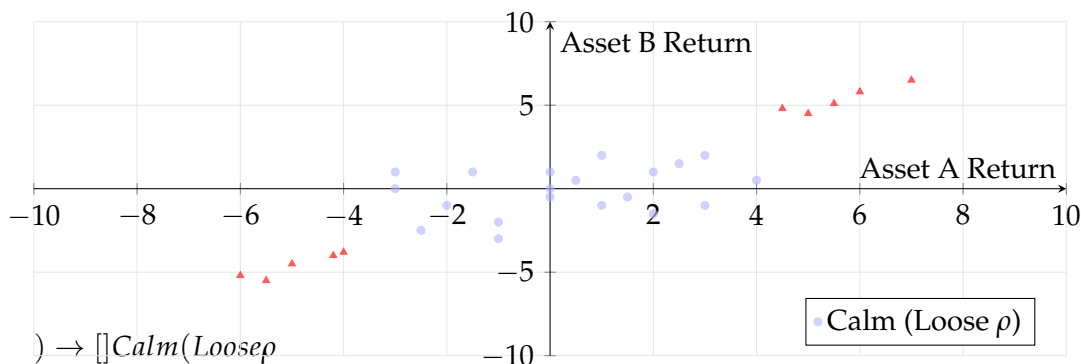


Figure 3: Correlation Regime Shift: Red points (Stress) align perfectly, unlike Blue points (Calm). Diversification vanishes.

6.2 Cross-Gamma PnL: The Surprise Term

Two-asset PnL expansion: -

Core Formula

$$\Delta V = \Delta_1 dS_1 + \Delta_2 dS_2 + \frac{1}{2} \Gamma_{11} dS_1^2 + \frac{1}{2} \Gamma_{22} dS_2^2 + \underbrace{\Gamma_{12} dS_1 dS_2}_{\text{cross-gamma}}$$

Core Formula

$$\text{PnL} = \underbrace{\text{Gap}}_{\text{jump}} + \underbrace{\text{Slippage}}_{\text{liquidity}} + \underbrace{\text{Cross-Gamma}}_{\text{correlation}} + \underbrace{\text{Model Failure}}_{\text{dynamics wrong}}$$

All terms are orders of magnitude larger than normal Greeks.

9.3 Which Hedges Stopped Working First?

Hedge failure sequence: 1. Correlation hedges (first to fail) 2. Vega hedges (surface freezes) 3. Delta hedges (liquidity vanishes) 4. Gamma hedges (cost explodes)

Diagnosis: The first hedge to fail tells you which assumption broke first.

Core Formula

$$\text{Failed Hedge} \implies \text{Broken Assumption}$$

Mapping: - Correlation hedge failed $\rightarrow$ correlation $\rightarrow$ 1 - Vega hedge failed $\rightarrow$ vol surface stuck - Delta hedge failed $\rightarrow$ liquidity gap - Gamma hedge failed $\rightarrow$ discrete hedging cost

"Stress PnL Triage: First Failure = Root Cause"

Desk practice: During crisis, log which hedge you couldn't execute first. That's the assumption to fix.

Interview: "In March 2020, correlation hedges failed first. Told me correlation matrices were useless. I cut basket notionals by 50% immediately."

9.4 Which PnL Component Flipped Sign?

Normal: - Delta PnL: positive if spot up, negative if down - Vega PnL: positive if vol up if long vega - Gamma PnL: negative if gamma short

Stress: - Delta PnL: positive **and** negative on same day (jumps up and down) - Vega PnL: sign flips as surface inverts - Gamma PnL: becomes positive for short gamma (gap risk premium)

Sign flip = dynamics changed.

Practitioner Insight

The Vol Inversion Signal:

In March 2020, short-dated vol traded above long-dated vol (inverted term structure). Vega PnL for long-dated options flipped sign vol up hurt long-dated vega. The model didn't have term structure dynamics. Stress revealed missing factor.

9.5 Which Assumption Failed Earliest?

Timeline of failure:

- T-3 days: Correlation matrix becomes unstable (eigenvalues spike)
- T-1 day: Skew flattens (RR compresses)
- Event day: Liquidity gap (bid-ask blows out)
- T+1 day: Vol surface re-calibrates (dirty PnL dominates)